When More Context Hurts: Evaluating the Impact of Irrelevant Information on LLM Reasoning Performance

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ABSTRACT

- Standard RAG and long-context systems assume more context = better outputs
- This paper tests whether irrelevant context *hurts* LLM reasoning
- Evaluates Gemma-3-1B across 3 task types under 3 prompt conditions
- Key finding: Irrelevant context never outperforms minimal or relevant context
- Chain-of-thought prompting rescued GSM8K accuracy from 4% → 46%
- Irrelevant context caused a 10 percentage point drop in commonsense reasoning

Results

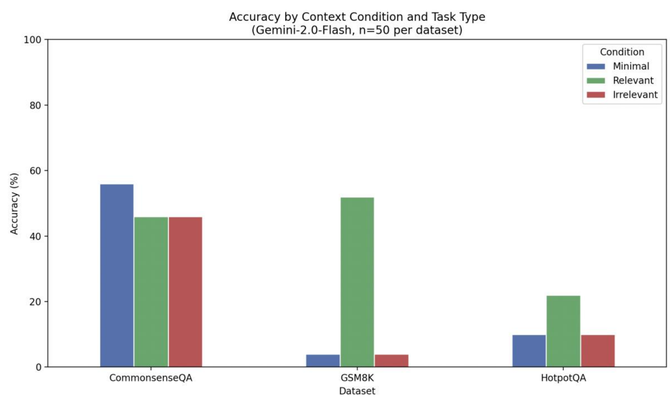

Figure 2: LLM reasoning accuracy under three prompt conditions across three task types (Gemma-3-1B, $n = 50$ per dataset).

DISCUSSION

Two mechanisms:

- **Attentional competition** — irrelevant tokens compete with relevant ones in self-attention
- **Heuristic over-reliance** — models learned during pretraining that "context is usually relevant," so they try to use it even when it isn't

EXPEREMENTAL SETUP

Model: Gemma-3-1B via Google AI Studio API

3 Datasets (50 samples each, seed=42):

- GSM8K :Arithmetic reasoning (math word problems)
- CommonsenseQA :Commonsense (5-choice multiple choice)
- HotpotQA : Multi-step reasoning (multi-hop QA)

Condition	What model sees?
Minimal	Question only
Relevant	Question + helpful context
Irrelevant	Question + unrelated factual noise

Irrelevant context pool: 5 sentences from unrelated domains (European history, geography, architecture, music history, physics) — factually correct but completely off-topic.

Dataset	Minimal (%)	Relevant (%)	Irrelevant (%)
CommonsenseQA	56.0	46.0	46.0
GSM8K	4.0	46.0	4.0
HotpotQA	10.0	22.0	10.0
Average	**23.3**	**38.0**	**20.0**

Table 3: Accuracy across datasets under minimal, relevant, and irrelevant context conditions. Relevant context improves arithmetic and multi-hop reasoning substantially, while irrelevant context generally fails to improve and may reduce performance relative to stronger baselines.

- Irrelevant context *never* beats minimal or relevant across any task
- GSM8K: chain-of-thought gave a **+42 percentage point** jump
- CommonsenseQA: irrelevant context caused a **−10 percentage point** drop vs. minimal
- Average irrelevant (20%) even underperforms minimal (23.3%)

ERROR ANALYSIS & LIMITATIONS

3 Failure Modes identified from 12 cases where irrelevant context flipped a correct answer wrong:

1. Answer Choice Shift (58% of failures — 7 cases) Model picks the right answer with no context, but switches to wrong answer when irrelevant text is added.
Example: "Where could you find hundreds of thousands of homes?" correct: D, with irrelevant Eiffel Tower fact → model picks A

2. Topic Misdirection (25% of failures — 3 cases) Model uses the irrelevant sentence as a topic cue and reasons about the wrong domain. *Example: Musical instrument question + Beethoven fact → model goes music-domain but still picks wrong answer*

3. Reasoning Chain Disruption (17% of failures — 2 cases) Multi-step reasoning gets interrupted mid-chain; model produces partial or off-topic answers. *Primarily observed in HotpotQA*

LIMITATIONS

- Only 50 samples per condition (pilot study, not statistically definitive)
- Single model (Gemma-3-1B) may not generalize to larger models
- English only
- Small distractor pool (5 sentences)
- No CoT tested in the irrelevant condition
- HotpotQA used substring matching instead of token-level F1

CONCLUSION AND FUTURE WORK

- Larger sample sizes with confidence intervals
- Test across multiple model sizes (GPT-4, etc.)
- Richer distractor types (topically adjacent, adversarial
- Multilingual settings
- Chain-of-thought + irrelevant context combination testing
- **Commonsense:** Most vulnerable to raw accuracy drop use strict relevance filtering
- **Arithmetic:** Needs structured chain-of-thought regardless of other context
- **Multi-step:** Sensitive to both irrelevant context disrupts intermediate reasoning steps

GITHUB LINK

github.com/haribrunda/llm-context-sensitivity